

2026 MIDDLE SCHOOL STEM CAMPS

From Trees to Tech

INSTRUCTOR GUIDE PACKET

Facilitation guides for every primary and backup activity.

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Mud Battery Bioelectric League

Turn a cup of mud into a living battery

At a glance: Can your team make mud generate electricity and defend your design with data? About 80 min; 4 teams.

MATERIALS

- 4 DIY soil microbial fuel cells: clear deli container, pond mud, two 4×4 in carbon-felt electrodes each (plus 1 spare set)
- measured sugar cubes as fuel (one per cell per run, plus spares)
- 4 multimeters, one per team, plus 2 spare (confirm battery type and that a battery is included)
- 100 kΩ resistors and alligator-clip lead wires, one set per cell
- nitrile gloves and a trash-bag drip liner per station
- labels and masking tape
- handwashing supplies

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which design change moved your readings the most, and how do you know?
- Why must the anode stay buried and the cathode stay in air?
- How is your weekly log like the work of an energy engineer?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage reading	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	100

SOURCES soil microbial fuel cell electrode and voltage references (carbon-felt anodes; terrestrial MFC literature on 300 to 600 mV output as the biofilm matures).

Forest Sensor Sprint

Site the smartest place for a campus tree sensor

At a glance: Find the smartest place to install a future campus tree sensor. About 80 min; 4 teams.

MATERIALS

- 4 clipboards plus spares
- 4 paper clinometers plus spares
- 4 combined temperature + humidity meters (one device, plus spares)
- 4 three-in-one soil meters (moisture, light, pH), one per team, plus spares
- 1 long tape measure (pre-mark a fixed standoff distance in prep)
- 4 route cards (one per team) plus 2 spares
- pencils

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: calibrate against the reference first, record every reading with its units, and plan a route with no backtracking.
- Calibrate together, then measure temperature, humidity, light, and soil moisture with units at each checkpoint and estimate the marked tree's height.
- Order the checkpoints to minimize backtracking, then recommend one sensor site and defend it with your own numbers.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which measurement most influenced your choice?
- Where did two nearby spots disagree, and why?
- What would change your recommendation in winter?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Data accuracy	30
Route efficiency	20
Site recommendation evidence	30
Teamwork and field conduct	20
Total	100

SOURCES GLOBE Observer Trees; NASA MyNASAData Urban Heat Island

Seed Dispersal Derby

Engineer a seed that escapes its parent tree

At a glance: Build a seed that escapes the parent tree without engines or throwing. About 80 min; 4 teams.

MATERIALS

- paper
- cardstock
- paper clips
- masking tape
- string
- fan or marked test lane
- a shared tape measure for distance
- timers

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
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- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which change improved hang time without losing distance?
- Why does spinning slow the fall?
- Which tree's strategy did your final seed copy?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Distance traveled	35
Hang time	25
Target landing	15
Biomimicry explanation	15
Redesign improvement	10
Total	100

SOURCES Science Buddies Seed Dispersal; Project Learning Tree

Xylem Pipeline Relay

Move water uphill the way a tree does

At a glance: Move water uphill like a tree, faster and cleaner than rival teams. About 80 min; 4 teams.

MATERIALS

- paper towels
- cotton balls
- cotton cloth strips
- cups
- food coloring
- trash bag table cover
- rulers
- elevation blocks

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which material moved water best, and why?
- How did slope change your delivery?
- Where does a real tree use this same effect?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Water delivered	40
Speed	20
Least spill	15
Design explanation	15
Redesign gain	10
Total	100

SOURCES Science Buddies; plant water transport references

Greenhouse Climate Controller

Match plants to the right climate zone

At a glance: Run the greenhouse controls for a plant that cannot complain, only wilt. About 80 min; 4 teams.

MATERIALS

- plant profile cards
- control dial boards
- clipboards
- dry erase markers
- tour clue sheets

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: match on what each plant needs, use the tour evidence, and change one placement at a time.
- Sort the plant cards by need, match each plant to a zone, and set temperature, humidity, and light on the dial board using the tour evidence.
- Where plants conflict in one zone, pick the setting your evidence best supports and note the tradeoff.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which plant was hardest to place, and why?
- What tradeoff did you have to accept?
- How do real growers handle conflicting needs?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct setting choices	30
Use of tour evidence	25
Explanation	20
Layout efficiency	15
Teamwork	10
Total	100

SOURCES Temple Ambler Greenhouse

Pinecone Weather Machine

A humidity sensor with no battery and no code

At a glance: Build a humidity sensor with no battery and no code. About 80 min; 4 teams.

MATERIALS

- dry pine cones
- index cards
- wood skewers
- spray bottles
- rulers, and a timer per team to time the response
- clear tape (packing or cellophane)
- paper fasteners

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does a two-layer strip bend when only one layer swells?
- What made your indicator faster?
- Where would a no-power humidity sensor be useful?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Response accuracy	35
Response speed	20
Readable output scale	20
Biomimicry explanation	15
Redesign quality	10
Total	100

SOURCES AskNature Pine Cones Strategy

Pollinator Network Draft and Build

Design a habitat that works all season

At a glance: Draft a pollinator habitat that works all season, not just on the prettiest day. About 80 min; 4 teams.

MATERIALS

- plant cards
- pollinator cards
- grid boards
- stickers
- markers
- season tokens

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: native plants and clumping; cover spring, summer, and fall; change one placement at a time.
- Draft and place the plant and pollinator cards on the grid, then check the season coverage.
- Find a gap or an unfed pollinator, fix one placement, and recheck the coverage.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Where was the biggest gap in your bloom calendar?
- Why does clumping help pollinators?
- What happens to the network if one season is missing?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Seasonal coverage	30
Pollinator fit	25
Native and clumping logic	20
Layout clarity	15
Defense	10
Total	100

SOURCES USDA Pollinators; US Forest Service pollinator guidance

Arboretum Eco-Quest

Solve outdoor ecology checkpoints

At a glance: Read the living collection and solve the checkpoints before time runs out. About 80 min; 4 teams.

MATERIALS

- clue cards
- route map and laminated adaptive evidence key per team; keep the staff route key for verified species names
- clipboards
- pencils
- evidence token bags
- optional QR codes

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Field method: stay together on approved paths, plan a no-backtrack route, record observable traits, and use verified sources for species names.
- Walk the route and use the clue cards and adaptive evidence key to record observable traits. Use a staff-verified numbered field tag, visible label, or route key for any species name, then collect the evidence token.
- Check that every answer cites an observed trait and that every species name uses a verified field tag, visible label, or staff route key before returning.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which clue was most reliable for telling trees apart?
- Where did a key send you wrong, and why?
- What makes a good piece of field evidence?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	100

SOURCES Temple Ambler Arboretum; Arboretum Explorer

Minecraft Tree World Resilience Cup

Design a climate-resilient landscape

At a glance: Design a landscape that survives stress because of how it is built, not luck. About 80 min; 4 teams.

MATERIALS

- Minecraft Education devices and template or paper grid kit
- rubric cards
- display timer

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which feature would help most in a real storm?
- Why does diversity add resilience?
- What in your build would fail first, and how would you fix it?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Resilience features	30
Biodiversity support	25
Realism	20
Explanation	15
Build polish	10
Total	100

SOURCES Minecraft Education Biodiversity

Tree Ring Climate Detective

Practice reading stylized rings as proxy evidence

At a glance: Use an authored ring-card code to build and defend a model history, then explain what real evidence would require. About 80 min; 4 teams.

MATERIALS

- printed ring cards
- wood cookies or images
- answer boards
- timers
- claim-evidence cards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Practice-card limits: read annual bands center to bark, apply only the authored code, cite exact bands, and distinguish the model from real cross-dated and calibrated evidence.
- Read the authored band sequences and assign model events under the printed practice-card code.
- Support each model claim with exact annual bands, synthesize a practice-card history, and state what real cross-dating and local calibration would require.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Which pattern was easiest to misread?
- How is a tree ring like and unlike a thermometer?
- What other natural records store past climate?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Practice-code inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	100

SOURCES NOAA NCEI Tree Ring product and paleoclimatology (<https://www.ncei.noaa.gov/products/paleoclimatology/tree-ring>); NOAA Climate.gov, How tree rings tell time and climate history (<https://www.climate.gov/news-features/blogs/beyond-data/how-tree-rings-tell-time-and-climate-history>); UCAR Center for Science Education, Tree Rings and Climate (<https://scied.ucar.edu/learning-zone/how-climate-works/tree-rings-and-climate>).

Lotus Leaf Surface Sprint

Engineer a self-cleaning, water-shedding surface

At a glance: Build a surface that sheds water and dirt the way a lotus leaf does. About 80 min; 4 teams.

MATERIALS

- wax paper
- cardstock
- sandpaper strips, and a candle or paraffin wax to coat a surface
- masking tape
- pipettes
- water
- pepper or cocoa residue
- ramps

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does a rough, waxy surface shed water better than a smooth one?
- What carried the dirt away?
- Where would a self-cleaning surface be useful?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Fast clean runoff	35
Least residue	25
Surface design explanation	20
Redesign gain	10
Craftsmanship	10
Total	100

SOURCES AskNature Lotus Leaf Self-Cleaning Surface

Leaf Stomata Microscope Detective

Compare standardized stomatal density without overclaiming water use

At a glance: Compare stomatal density with a standardized count and make a cautious, testable hypothesis. About 80 min; 4 teams.

MATERIALS

- leaf samples
- clear tape
- clear nail polish or prepared slides
- USB microscopes or magnifiers
- grid sheets
- gloves

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Standardize surface, preparation, magnification, field area, and counting rule; count 3 or more fields and report mean plus range.
- Prepare and count samples with the fixed method, then compare mean stomatal density and range.
- Make a cautious hypothesis and state that counts alone cannot rank actual water use; aperture, pore size, gas exchange, species, and conditions also matter.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why count several fields instead of one?
- Why can stomatal counts alone not determine a leaf's actual water use?
- Where might error sneak into your count?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Slide or peel quality	25
Standardized counting accuracy	30
Cautious evidence-backed hypothesis	25
Mean-and-range data table	10
Cleanup and care	10
Total	100

SOURCES California Academy of Sciences Stomata Printing; Rothamsted Bioimaging; Bertolino, Caine, and Gray (2019), PMCID PMC6414756; Lunn et al. (2024), PMCID PMC11565199; USDA Forest Service hardwood physiology research.

BACKUP ACTIVITIES

Ready alternates

Root Grip Erosion Rescue

Can roots hold a slope together in a storm?

At a glance: Can roots hold a slope together during a storm? About 60 min; 4 teams.

MATERIALS

- Trays
- Soil
- String or yarn roots
- Craft sticks
- Watering cups

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why did the rooted slope hold more soil?
- How would deeper roots change the result?
- Where is this used to protect real hillsides?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Soil retained	40
Runoff control	25
Design explanation	20
Cleanup	15
Total	100

SOURCES EPA green infrastructure and erosion lessons

Tree Height Triangulation Shootout

Measure a tree without climbing it

At a glance: Measure a tree without climbing it. About 60 min; 4 teams.

MATERIALS

- Paper clinometers
- String and washers
- Rulers and tape measures
- Clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- What was your biggest source of error?
- Why does eye height matter in the calculation?
- Where is remote height measurement used in the field?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	100

SOURCES GLOBE Trees Activities

Urban Heat Shade Dash

Find the coolest route through a hot campus

At a glance: Find the coolest route through a hot campus. About 60 min; 4 teams.

MATERIALS

- Infrared (non-contact) thermometers
- Campus maps
- Clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- How big was the sun-to-shade temperature gap?
- What surface was hottest, and why?
- How do cities use shade to cool streets?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Cool route design	35
Data quality	25
Reasoning	20
Communication	20
Total	100

SOURCES NASA MyNASADData Urban Heat Island

Photosynthesis Float-Off Playoffs

Make leaf disks float by making oxygen

At a glance: Make leaf disks float by producing oxygen. About 60 min; 4 teams.

MATERIALS

- Spinach leaves
- Baking soda solution
- Cups
- Oral syringes, no needle
- Timers

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does oxygen make the disks float?
- Which variable changed the rate most?
- How is this evidence of photosynthesis?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Controlled-variable design	25
Data table	20
Explanation	10
Cleanup	10
Total	100

SOURCES Exploratorium Photosynthetic Floatation